① 4.1 — 7

$Ax = b$ $\qquad\qquad$ $A^T y = 0$

$y^T b = 1$

$$\left[\begin{array}{ccc|c} 1 & -1 & 0 & 1 \\ 0 & 1 & -1 & 1 \\ 1 & 0 & -1 & 1 \end{array}\right]$$

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\lambda \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} \qquad y \rightarrow \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$$

② 4.1 - 9

$A^T A x = 0$ then $Ax = 0$

i) column space

ii) orthogonal

③ 4.1 - 31

in matlab

$N =$ null$(A)$ $\to$ basis for $N(A)$

$B =$ null$(N^T)$ In matlab

$N^T$ ! $\quad N'' = N^T$

$$N = \begin{bmatrix} | & | & & | \\ n_1 & n_2 & \cdots & n_k \\ | & | & & | \end{bmatrix}$$

colspace $(N)$ = nullspace of $(A)$

rowspace $(N^T)$ = nullspace of $A$

$$\begin{bmatrix} — n_1 — \\ \vdots \\ — n_k — \end{bmatrix} = N^T$$

$\rightsquigarrow$ nullspace of this will be all vectors $\perp$ to rowspace of $N^T$

All vectors $\perp$ null space of $A$

$\hookrightarrow$ row space of $A$

④ 4.1 - 32

$r, n, c, l \rightarrow$ non-zero vector

a) rank $> 0$ rank $= 1$

rank $< 2$

$C(A^T) \perp N(A)$ $r^T n = 0$

$C(A) \perp N(A^T)$ $c^T l = 0$

b) $A = \begin{bmatrix} 1 & 9 \\ 3 & 27 \end{bmatrix}$ $c = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$

$\begin{bmatrix} 1 & 9 \\ 0 & 0 \end{bmatrix}$ $n = \begin{bmatrix} -9 \\ 1 \end{bmatrix}$

$\begin{bmatrix} 1 & 3 \\ 0 & 0 \end{bmatrix}$ $r = \begin{bmatrix} 1 \\ 9 \end{bmatrix}$

$l = \begin{bmatrix} -3 \\ 1 \end{bmatrix}$

(6) 4.1 - 33

$r_1, r_2$ , $n_1, n_2$ , $c_1, c_2$ , $l_1, l_2$ $\mathbb{R}^4$

$R = \begin{bmatrix} r_1 & r_2 \end{bmatrix}_{4\times 2}$ $N = \begin{bmatrix} n_1 & n_2 \end{bmatrix}_{4\times 2}$ $C = \begin{bmatrix} c_1 & c_2 \end{bmatrix}_{4\times 2}$

$L = \begin{bmatrix} l_1 & l_2 \end{bmatrix}_{4\times 2}$

a) $r_1^T n_1 = 0$ $R^T N = 0$ $\boxed{\text{rank} = 2}$

$r_2^T n_2 = 0$

$c_1^T l_1 = 0$ $C^T L = 0$

$c_2^T l_2 = 0$

All non zero, if zero vector then we would need 1, 2, 3 → in the list

b) Possible matrix → Any $n \times n$ rank 2 matrix

$$\begin{bmatrix} 1 & 0 & 2 & 3 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

⑥ 4.2 - 13

$A \to 4\times4 = I$

$$4\times3 \quad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \quad \xleftarrow{\text{project}} \; b = (1,2,3,4)$$

$$\Rightarrow \quad P = A\,(A^T A)^{-1} A^T$$

Since A has orthonormal columns

$\boxed{A^T A = I}$

$$P = A A^T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

4x4 matrix

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \qquad p = (1,2,3,0)$$

7 4.2-16

$$\begin{bmatrix} 1 & 1 \\ 2 & 0 \\ -1 & 1 \end{bmatrix} \xleftarrow{\text{project}} b = \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}$$

orthogonal columns

$$A^T A = \begin{bmatrix} 1 & 2 & -1 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 2 & 0 \\ -1 & 1 \end{bmatrix}$$

$n \times m$ $m \times 1$

$$\begin{bmatrix} 1+4+1 & 0 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 6 & 0 \\ 0 & 2 \end{bmatrix}$$

$$A^T A \hat{x} = A^T b \qquad A^T b = \begin{bmatrix} 1 & 2 & -1 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}$$

$$2 \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix} \hat{x} = 3 \begin{bmatrix} 1 \\ 1 \end{bmatrix} \qquad A^T b = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 0 \\ 0 & 2/3 \end{bmatrix} \hat{x} = \begin{bmatrix} 1/2 \\ 3/2 \end{bmatrix}$$

8) <u>4.2-17</u>

if $P^2 = P$ show $(I-P)^2 = I-P$

\# $(I-P)^2 = I^2 + P^2 - 2IP$

$= I + P - 2P = I - P$

$\Rightarrow (I-P)^2 = I - P$

\# P projects onto the column space of A, I-P projects onto left null space.

$p = Pb$

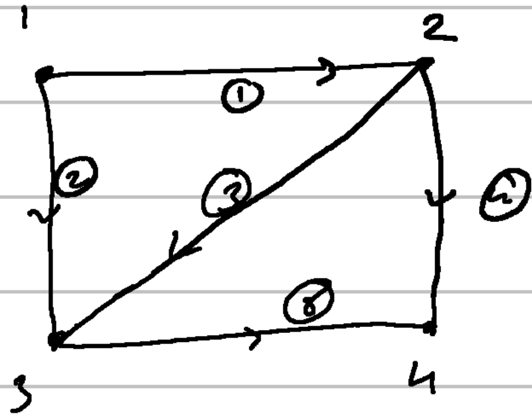
$b = p + e$

9 8.2-13

prev page → 218

$c_1 = c_2 = 2$

$c_3 = c_4 = c_5 = 3$




$$C = \begin{bmatrix} 2 \\ 2 \\ 3 \\ 3 \\ 3 \end{bmatrix}$$

$$A = \begin{array}{cccc} 1 & 2 & 3 & 4 \end{array}$$
$$\begin{bmatrix} -1 & 1 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & -1 & 0 & 1 \\ 0 & 0 & -1 & 1 \end{bmatrix}$$

$5\times4$

$4\times5 \quad 5\times1$

$4\times1 \quad 5\times4$

$\boxed{A^T C A x = f}$ used matlab!

10) 8.2 - 12

nodes $\rightarrow$ 9
edges $\rightarrow$ 12

$\boxed{\text{loops} = 4}$

a) tree with rank = 8

$\boxed{12 - 4 = 8 \text{ edges}}$

b) $A^T y = f$

$f \in C(A^T)$ which is orthogonal to $N(A)$

$N(A) \rightarrow \dim = 1$

For connected graphs

$N(A) = \lambda [1, 1 - - -]^T$

$f \rightarrow$ sum of all components $= 0$

(11) 4.2 - 30

a) $A = \begin{bmatrix} 3 & 6 & 6 \\ 4 & 0 & 8 \end{bmatrix}$ rank = 1

$a = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$ $P_c = \dfrac{a a^T}{a^T a} =$

$$P = \frac{\begin{bmatrix} 3 \\ 4 \end{bmatrix} \begin{bmatrix} 3 & 4 \end{bmatrix}}{\begin{bmatrix} 3 & 4 \end{bmatrix} \begin{bmatrix} 3 \\ 4 \end{bmatrix}}$$

$$\boxed{= \frac{1}{25} \begin{bmatrix} 9 & 12 \\ 12 & 16 \end{bmatrix}}$$

b) $a = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} \begin{bmatrix} 1 & 2 & 2 \end{bmatrix} \dfrac{1}{9}$ $= \dfrac{1}{9} \begin{bmatrix} 1 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 4 \end{bmatrix}$

(13) 4.2 - 31

$\mathbb{R}^m$ $b, p$

$p = c_1 a_1 \ldots c_n a_n$

$(b-p)^T a_i = 0$

(13) 4.2-34

$A \to r$ independent column $\to m \times r$
$B \to r$ independent rows $\to r \times k$

then $AB$ is invertible

\# $A^T A$ ✓ invertible $(r \times r)$ rank $r$
$B B^T$ ✓ invertible $(r \times r)$ rank $r$

$\underbrace{A^T A}_{\text{rank } r} \underbrace{B B^T}_{\text{rank } r}$   $AB \to \boxed{\text{rank } r}$

There is a problem in solution!